Recursion

The process in which a function calls itself is called recursion and the function that calls itself is called a recursive function.

C program to find the factorial of a number *using recursion*

```
#include<stdio.h>
int fact(int);
void main( )
{
   int n,x;
   printf("\nEnter the number”);
   scanf("%d",&n);
   x=fact(n);
   printf("\nThe factorial is %d",x);
}
int fact(int m)
{
   if(m==1)
      return 1;
   else
      return (m*fact(m-1));
}
```

Output

Enter the number 6

The factorial is 720

C program to find the nth term of Fibonacci series *using recursion*

```
#include<stdio.h>
int fib(int);
void main( )
{
```

```
    int n,a;
    printf("Enter the nth number in fibonacci series: ");
    scanf("%d",&n);
    a=fib(n);
    printf("\nThe %d-th number in fibonacci series is %d ", n, a);
  }
int fib(int m)
{
    if(m==1)
    {
            return 0;
    }
    else if(m==2)
    {
            return 1;
    }
    else
    {
            return(fib(m - 1) + fib(m - 2));
    }
}
```

Output

Enter the nth number in fibonacci series: 5

The 5-th number in fibonacci series is 3

C program to display the Fibonacci series upto n terms *using recursion*

```
#include<stdio.h>
int fibo(int);
void main()
{
   int n, i=0, c;
```

```
  printf("\nEnter the no of terms:");
  scanf("%d",&n);
  printf("Fibonacci series is\n");
  for(c=1;c<=n;c++)
  {
    printf("%d  ", fibo(i));
    i++;
  }
}
int fibo(int n)
{
  if (n==0)
    return 0;
  else if (n==1)
    return 1;
  else
    return ( fibo(n-1) + fibo(n-2) );
}
```

Output

Enter the no of terms: 6

Fibonacci series is

0 1 1 2 3 5

C program to find the GCD of 2 numbers *using recursion*

```
#include<stdio.h>
int gcd(int, int);
void main( )
{
   int a, b, c;
   printf("Enter the 2 numbers\n ");
   scanf("%d%d", &a, &b);
```

```
    c=gcd(a,b);
    printf("G.C.D is %d",c);
}
int gcd(int x, int y)
{
     if (y != 0)
     {
              return gcd(y, x%y);
     }
     else
     {
              return x;
      }
}
```

Output

Enter the 2 numbers

21 9

G.C.D is 3